Request and prediction

MUX

Fetch PC

Request gate
Ready / not busy

I-Cache / TCM

Outstanding bit
Accept: set
Response: clear

Branch
training

BTB / BHT / RAS

Accepted context: PC / target / flags

Response and buffering

Live response
Data / PC /
prediction

MUX

Valid gate
(response OR skid)
AND NOT drop

Predecode

Instruction FIFO

Capture
valid & !ready

Skid data + valid
132-bit payload

Drop window
branch_q OR branch_d_q

Redirect and flush

Backend redirect
+ raw FIFO flush

Redirect PC / valid
branch_q

Delay bit
branch_d_q